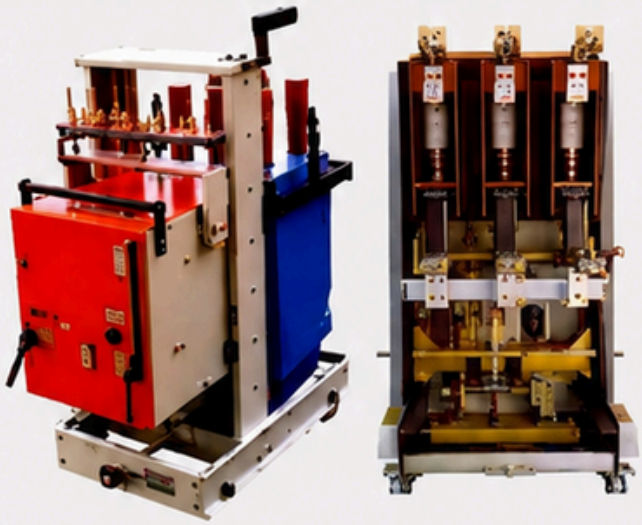
BMC

EXCELLENCE REDEFINED ®

Since 1986

RETROFIT SOLUTION

- One-third the price of a new installation
- No demolition and construction investment
- No substation redesign work
- Minimum down time



BMC ELECTROPLAST COMPANY PROFILE

BMC ELECTROPLAST was founded in 1986 by two young engineers, Late Mr. Santanu Mitra and Mr. Subodh Banerjee after completing their B E from REC, DURGAPUR (Presently NIT, DURGAPUR) BMC is currently led by a group of four engineer Directors

Mr. Subodh Banerjee (B.E. - ELECTRICAL)
Mr. Rajiv Bhargava (B.E. - MECHANICAL)
Mr. Siddhartha Mitra (B.E. - CHEMICAL)
Mr. Girish V Magre (P.G. - ELECTRICAL VACUUM TECHNOLOGY)

BMC has a workforce of 150 plus employees comprising of graduate engineers, technical supervisors skilled workmen engaged in innovation technology enhancement of power systems equipment.

We are an Integrated Management System certified company by DNV GL.

BMC has two of their manufacturing units in Kolkata, West Bengal and one in Aurangabad, Maharashtra.

Head Office & Works-01 at Joka with a total area of 60 000 sq ft and a covered area of 30 000 sq ft along with a two storey office facility.

Works-02 Functioning as an assembly line at Behala Industrial Estate, Behala Chowrasta with a cover area of around 4 500 sq ft.

The manufacturing unit Works 03 at Aurangabad, Maharashtra with a total area of 70 000 sq ft and a covered area of 35 000 sq ft along with a two stories office facility and reserved for green space within the factory premises.

BMC is one of the leading manufacturers of medium voltage circuit breakers and it's major components for more than three decades having set up in Kolkata & Aurangabad.

BMC
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MILESTONES

Some of our pioneering works over the last three and a half decades being:

- First in India to offer technology for VCB/VI design to Indian manufacturers.
- First in Asia to offer technology for VCB/VI design to European and Overseas manufacturers.
- Pioneers in Epoxy Cast Vacuum Interrupters (Embedded Poles) for Vacuum contactors world wide.
- Unique compact design of MV Vacuum Contactors.
- First in Indian SME sector to get accredited as "R&D Centre" in switchgear field by DSIR (India).
- Representatives as "Technical/Marketing Consultants" for High Power Lab in Korea, Vacuum furnace manufacturer from Germany & component manufacturer for VI (single window solution for vacuum switchgears).
- Experience of designing and selling of retrofit VCBs of over 25 versions. Experience of self designing/manufacturing of custom built epoxy cast dry type instrument transformers (CT & PT) and other epoxy components for both Indoor and Outdoor switchgear applications.
- Being the first MSME in India to develop economic, indigenous, and reliable medium voltage Vacuum Circuit Breakers & Vacuum Contactors for Industry, bringing down costs by 30 percent.
- Developing Import substitutes developed for Electrical Power Distribution Sector and Indian Railways.
- Achieving accreditation as Global Suppliers for Cast Resin Instrument Transformers for major OEMs in India, South East Asia & Middle East.
- Over10+patents/ design registrations on Switchgears/ Switchgear components.
- Unique low cost Compact Sub Station design for Windmills in India

CERTIFICATIONS

First SME to be accredited as an "R&D" center for medium voltage switchgears by

DSIR (Department of Scientific and Industrial Research)

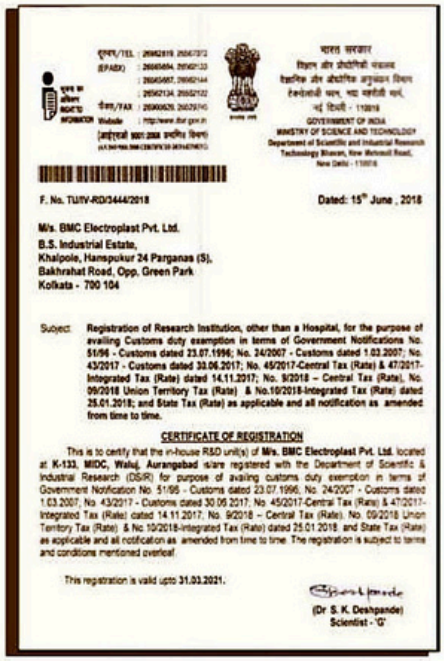
TÜV NORD

CERTIFICATE

Management system as per
ISO 14001 : 2015

BMC ELECTROPLAST PVT. LTD.
K 133, MIDC, Waluj,
Aurangabad - 431 136,
Maharashtra,
India

Design and Manufacture of Epoxy & Non Epoxy Cast Instrument Transformers for Medium Voltage Application. Manufacture of Epoxy Cast, BMC/DMC Compression Moulding Components. Design, Manufacture and Testing of Low & Medium Voltage Vacuum Contactors and Vacuum Circuit Breakers



GOVERNMENT OF INDIA
MINISTRY OF SCIENCE AND TECHNOLOGY
Department of Scientific and Industrial Research
Technology Bhawan, New Mehrauli Road,
New Delhi - 110016

F. No. TU/IV-RD/3444/2018 Dated: 15th June , 2018

M/s. BMC Electroplast Pvt. Ltd.
B.S. Industrial Estate,
Khalpole, Hanspukur 24 Parganas (S),
Bakhrahat Road, Opp. Green Park
Kolkata - 700 104

Subject: Registration of Research Institution, other than a Hospital, for the purpose of availing Customs duty exemption in terms of Government Notifications No. 51/96 - Customs dated 23.07.1996; No. 24/2007 - Customs dated 1.03.2007; No. 43/2017 - Customs dated 30.06.2017; No. 45/2017-Central Tax (Rate) & 47/2017-Integrated Tax (Rate) dated 14.11.2017; No. 9/2018 – Central Tax (Rate), No. 09/2018 Union Territory Tax (Rate) & No.10/2018-Integrated Tax (Rate) dated 25.01.2018; and State Tax (Rate) as applicable and all notification as amended from time to time.

CERTIFICATE OF REGISTRATION

This is to certify that the in-house R&D unit(s) of M/s. BMC Electroplast Pvt. Ltd. located at K-133, MIDC, Waluj, Aurangabad is/are registered with the Department of Scientific & Industrial Research (DSIR) for purpose of availing customs duty exemption in terms of Government Notification No. 51/96 - Customs dated 23.07.1996, No. 24/2007 - Customs dated 1.03.2007, No. 43/2017 - Customs dated 30.06.2017, No. 45/2017-Central Tax (Rate) & 47/2017-Integrated Tax (Rate) dated 14.11.2017, No. 9/2018 – Central Tax (Rate), No. 09/2018 Union Territory Tax (Rate) & No.10/2018-Integrated Tax (Rate) dated 25.01.2018 and State Tax (Rate) as applicable and all notification as amended from time to time. The registration is subject to terms and conditions mentioned overleaf.

This registration is valid upto 31.03.2021.

(Dr S. K. Deshpande)
Scientist - 'G'

IMS Certifications of Quality Compliance Management, Environment Compliance Management and Occupational Health and Safety Compliance

QCS MANAGEMENT PVT. LTD.
Certificate of Registration
ISO 9001:2015 (Quality Management System)
BMC ELECTROPLAST PVT. LTD.

QCS MANAGEMENT PVT. LTD.
Certificate of Registration
ISO 14001:2015 (Environmental Management System)
BMC ELECTROPLAST PVT. LTD.

QCS MANAGEMENT PVT. LTD.
Certificate of Registration
ISO 45001:2018 (Occupational Health & Safety Management System)
BMC ELECTROPLAST PVT. LTD.

BMC
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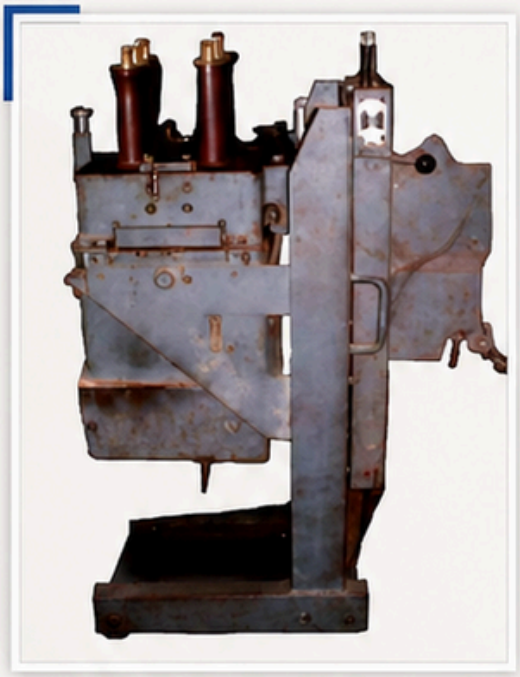
Since 1986

BMC BACKGROUND

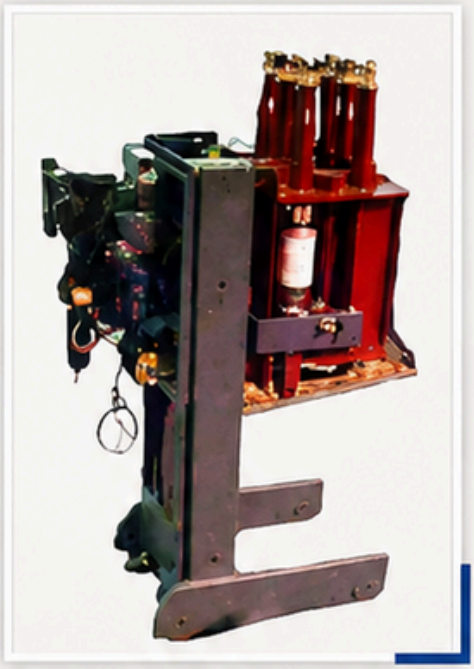
BMC HAS A BACKGROUND OF RETROFITTING OF VARIOUS OCBs AND VCBs OF OBSOLETE BRAND FOR NEARLY TWO DECADES AND HAS SUCCESSFULLY COMMISSIONED MORE THAN 2000 OCBs & VCBs OF 25+ DIFFERENT MAKES, ALL OVER THE COUNTRY AND ABROAD.

WHAT IS RETROFIT



Oil /SF6 circuit breakers

Retrofit



Vacuum circuit breakers

- Upgradation of Technology
- No Change in Panel
- Existing Breaker Remains as Spare

This method of introducing technology is called Retrofit

WHY RETROFIT

- **NEW LEASE OF LIFE FOR YOUR SWITCHGEAR**

Wear and tear occurs mainly in moving & arc quenching parts.

- **ALL THE ADVANTAGES OF VACUUM TECHNOLOGY COME AS BONUS VIZ:**

Very high reliability, safety, huge number of unattended interruptions, wide variety of applications such as motor, transformer & capacitor switching, etc.

- **NO ALTERATION IN THE EXISTING PANEL**

The existing breaker can be used as a spare one, hence doubling the reliability.

- **MAINTENANCE-FREE OPERATIONS**

Less downtime (can be huge savings).

- **REPLACEMENT TIME: 6-8 HRS**

Means savings for those who cannot afford shutdown.

- **NEW CIVIL WORKS OR CABLING GETS ELIMINATED (COSTS & TIME)**

Huge savings.

- **DESIRABLE IN CASE SPARES OF OLD BREAKERS ARE NOT AVAILABLE VIZ:**

Imported breakers.

- **MANUAL OPERATION BREAKERS CAN BE CONVERTED TO MOTORISED VERSIONS**

- **COST EFFECTIVENESS**

1. Reduced capital costs
2. No erection & commissioning cost
3. No civil works or cabling

Types Of Retrofitting Solutions With BMC

Off late we had concentrated on standardization of retrofits of different makes. We have successfully developed two basic modules as follows:

- **Vertical isolation module which has following common areas:**
  - **Mechanism**
  - **Drive Arrangement**
  - **Carriage Configuration**
  - **Inter lock arrangement**
  - **Isolating contact**

On case to case basis, altering only, the Earthing arrangement, the Guide Arrangement and the Shutter Opening arrangement would be necessary to match the existing panel

- **The horizontal isolation module, again has the similar common, features as above.**

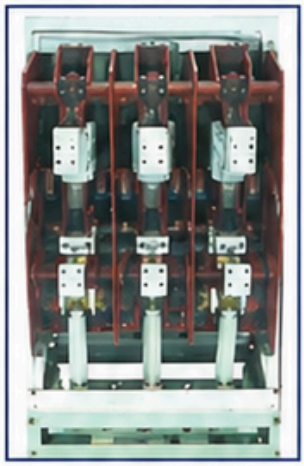
Our Experience

Till date we have already commissioned more than 3000 no.s of retrofit VCBs to various utilities and end users. The type of the various retrofit VCBs are as follows:

TYPE OF BREAKER	RATING OF THE BREAKER		
CONTACTOR MAKE	11kV, 630A, 25kA	11kV, 1250A, 20-25kA	
EAEG - AEGIS	11kV, 630A, 25kA		
BECOOL - BECO	11kV, 630A, 25kA	11kV, 1250A, 20kA	
BCH - BCHP	11kV, 630A, 25kA	11kV, 800A, 25kA	11kV, 1250A, 20kA
BCL - BCCB	11kV, 630A, 25kA		
BBC - BBCB	11kV, 630A, 25kA	11kV, 1250A, 20-25kA	
HYUN - HYNCH	11kV, 630A, 20-25kA	11kV, 1250A, 20-25kA	
KEC - KOCO	11kV, 630A, 25kA		
KHZ - KVCB	11kV, 630A, 20-25kA	11kV, 1250A, 20-25kA	
KEL - KOCH	11kV, 630A, 25kA	11kV, 1250A, 20kA	
KWH Japan - M600K	11kV, 630A, 25kA		
SCHNEIDER	11kV, 630A, 20-25kA	11kV, 1250A, 20-25kA	
SEMENS/SAFT/BHEL/ABB - BD/SD	11kV, 630A, 25kA		
STRÖMBERG/ HYUNDAI - UEE/CN	11kV, 630A, 25kA		
CROMPTON - NGC3	11kV, 400A, 25kA		
SIEMENS - 3AC/3AH	11kV, 630A, 20-25kA	11kV, 1250A, 20-25kA	
UNILOK - DOC03	11kV, 630A, 25kA		
GEC AL KHAZ - 3D-2/6	6.6kV, 800A, 18-25kA (Suitable for 20kA)		
CR CORPORATION - VC-1	11kV, 400A, 9kA (Suitable for 20kA)		

Retrofit Solution for Horizontal Isolation Breakers

NGEF Retrofit Solution



RETROFIT

Specifications

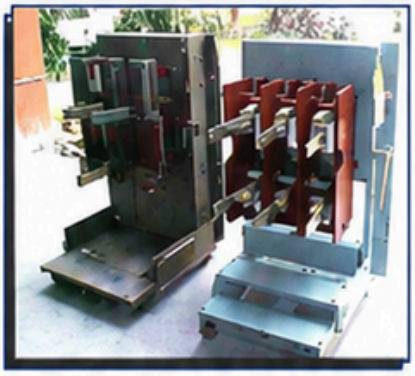
Type : SF6
Model : SFASG
Technology - AEG
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

END USER - Grasim, Nagda

NGEF Retrofit solution

Specifications

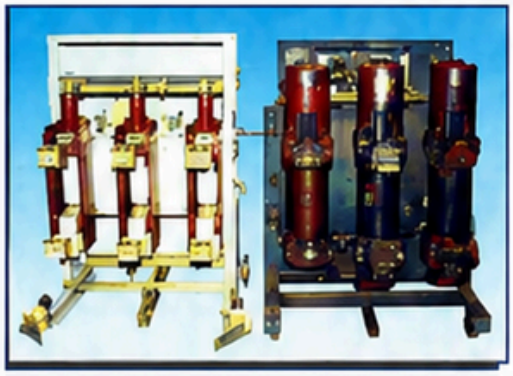
Type : MOCB
Model : GR256
Technology - AEG
Thermal rating - 630A
S.C.current - 20kA
System voltage - 6.6Kv



Comparative photo of original & retrofit NGEF (MOCB)

END USER - Mandovi Pellet, Goa

KIRLOSKAR Retrofit Solution



Specifications

Type : MOCB
Technology - ASEA
Thermal rating - 1600A
S.C.current - 25kA
System voltage - 11Kv

END USER - Delhi Vidyut Board

JYOTI Retrofit solution

RETROFIT
FRONT VIEW

Specifications

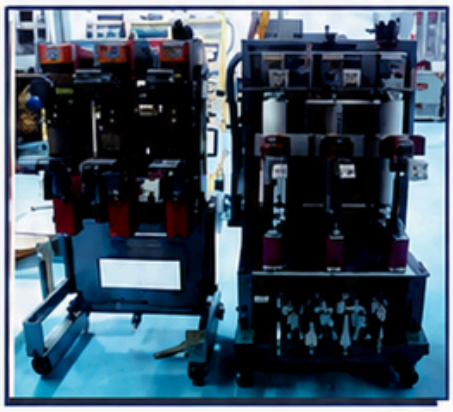
Type : MOCB
Model : OZ2/MT1
Technology : Calor Emag
Thermal rating - 800A
S.C.current - 25kA
System voltage - 6.6Kv

Rear view

END USER - Gujarat Electricity Board

ASEA Retrofit solution

MOCB TO VACUUM CONTACTOR

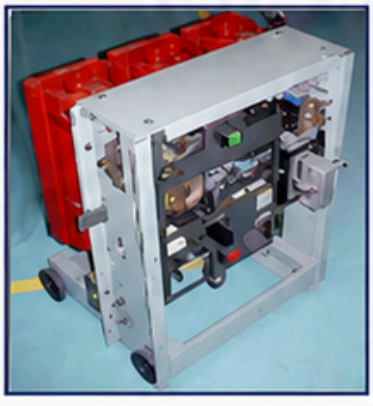


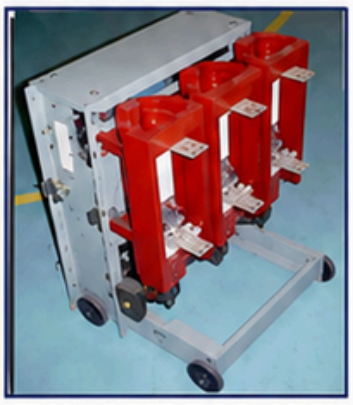
Specifications

Type : MOCB
Model : 12/1226
Thermal rating - 400A
System voltage - 6.6Kv

END USER - Vikram Cement

Retrofit solution Field switches for mining application





Retrofitted field switches

WESTING HOUSE Retrofit solution

ORIGINAL BREAKER

Specifications

Type : ACB
Model : DH P250
Thermal rating - 1250A
S.C.Current - 40kA
System voltage - 3.3Kv

RETROFIT SIDE VIEW

END USER - National Thermal Power Corporation, Talcher

Romanian make Retrofit Solution

OLD RETROFIT

Specifications

Type : BOCB
Thermal rating - 1250A
S.C.current - 18.4kA
System voltage - 11Kv

END USER - Indian Oil Corporation Ltd., Guwahati

STROMBERG Retrofit Solution

Specifications

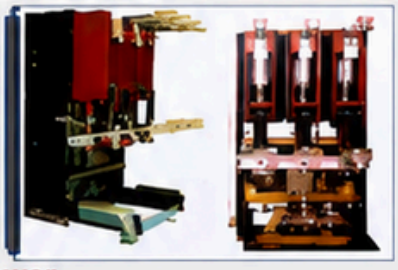
Type : BOCB
Model : OSAM12D
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

Bengaluru
OLD
Stromberg Breaker

RETROFITTED
Stromberg Breaker

END USER - Bosch, Bangalore

RETROFIT SOLUTION FOR NGEF MOCB



CUSTOMER: India Power
(Formerly Dishergarh Power Supply Company Limited)

TYPES OF RETROFITS

OLD

NEW

Specifications

Type : VCB

Thermal rating - 400A

S.C.current - 20kA

System voltage - 11Kv

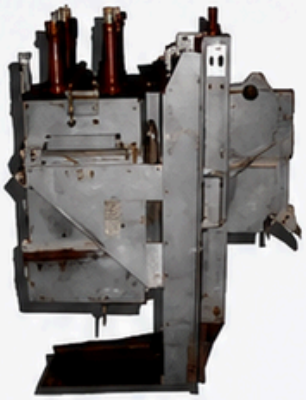
END USER -

Druk Green Power Corporation Limited, Bhutan

Retrofit Solution for Vertical Isolation Breakers

BIECCO Retrofit Solution

STD. MECHANISM



BIECCO BOCB

Specifications

Type : BOCB
Model : PDA/AG 16
Technology : Johnson Phillip
Thermal rating - 800A
S.C.current - 13.1kA
System voltage - 11Kv

RETROFIT VCB

END USER - West Coast Paper mills, Bangalore

MEI Retrofit Solution

RETROFIT
SIDE VIEW

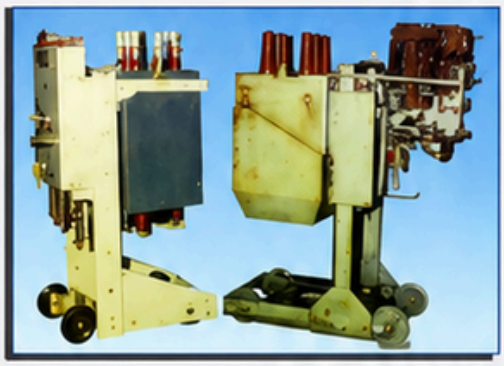
Specifications

Type : BOCB
Model : AC 1
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

FRONT VIEW

END USER - Lucas TVS, Chennai

CGL Retrofit Solution



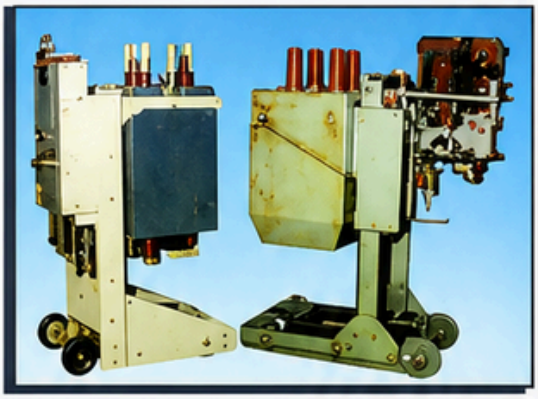
Specifications

Type : BOCB
Model : ALAX
Technology : Crompton Parkinson
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

Comparative photo of original & retrofitted CGL

END USER - TVS, Chennai

CGL Retrofit Solution



Specifications

Type : BOCB
Model : ALAX
Technology : Crompton Parkinson
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

Comparative photo of original & retrofitted CGL

END USER - TVS, Chennai

BHEL Retrofit Solution

Specifications

Type : BOCB
Model : BVP 3 & 4
Technology : English Electric
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

OLD

RETROFIT

Comparative photo of original & retrofitted BHEL

END USER - Bharat Heavy Electricals Limited, Trichy Tamilnadu

RBL Retrofit Solution

OLD

RETROFIT

Specifications

Type : BOCB
Model : LMT7X2/2X6/MO
Technology : Reyrolle
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

Comparative photo of original & retrofitted RBL

END USER - Bharat Electronics Ltd, Bangalore

ECE Retrofit Solution

Specifications

Type: BOCB
Model : HOP 5
Thermal rating - 800A
S.C.current - 18.4kA
System voltage - 11Kv

OLD

RETROFIT

Comparative photo of original& retrofitted ECE

END USER - West Bengal State Electricity Distribution Company

Southern Switchgear Retrofit Solution

OLD BOCB

RETROFIT

Specifications

Type: BOCB
Model : R4/1
Technology:Brush
Thermal rating - 400/800A
S.C.current - 26.2kA
System voltage - 11Kv

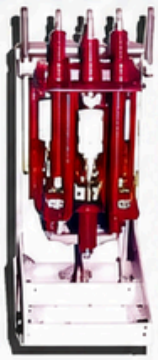
END USER - Ispat Industries, Nagpur

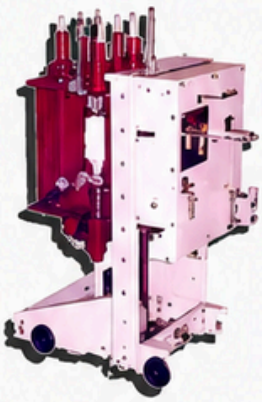
RETROFIT SOLUTION FOR
CGL BOCB

CUSTOMER: Brihanmumbal Electricity Supply and Transport

RETROFIT SOLUTION FOR BIECCO BOCB

RETROFIT SOLUTION FOR AECO BOCB



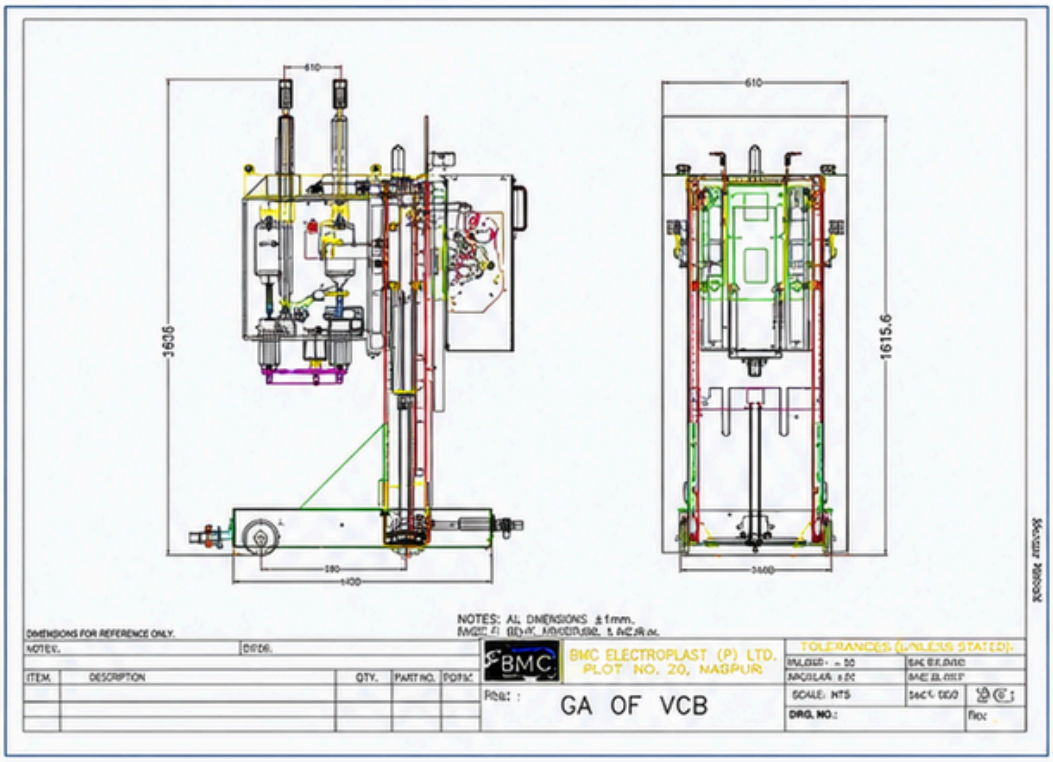


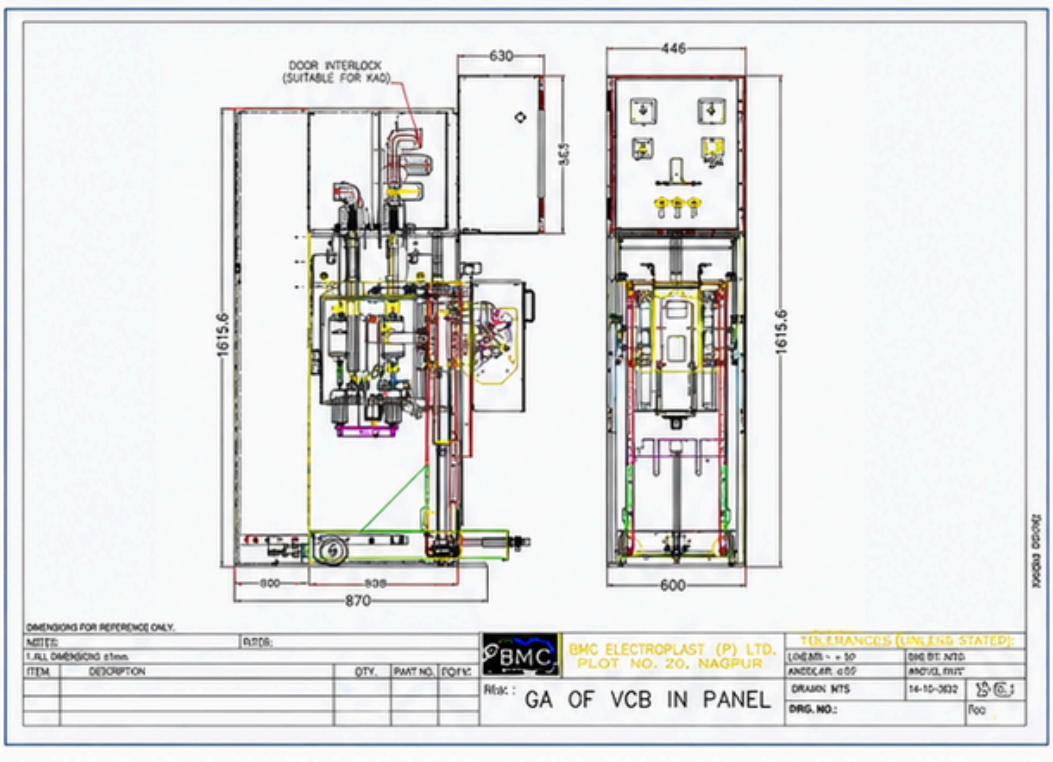
CUSTOMER: PRIVATE SECTOR INDUSTRY

RETROFIT SOLUTION FOR BVP17 BOCB/MMX

RETROFIT SOLUTION FOR KAD BOCB

RETROFIT SOLUTION FOR A GERMAN MAKE BOCB, DULY TYPE TESTED





BMC ELECTROPLAST PVT. LTD.

Reg. Office & Works1:130/22, B.S. Industrial Estate, Hanspukur, Bakhrahat Road, Kolakta - 700104(W.B.) India
Works 2 : K-133, Waluj, Aurangabad-431136(Maharastra), India

+91 93313 32975 // +91 87775 87692
: skb@bmcepl.col, siddhartha@bmcepl.com, bhargava@bmcepl.com, samrat@bmcepl.com, saptarshi@bmcepl.com
connect@bmcepl.com

BMC
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Since 1986

TECHNOLOGY CENTRE